

VELMÈRE BASIC AUDIT REPORT

Tornado Cash Router (0xd90e2f925da726b50c4ed8d0fb90ad053324f31b)

VELMERE SECURITY AUDIT REPORT: TORNADO CASH ROUTER

Network: Ethereum Mainnet (Chain ID: 1)

Report ID: undefined | Tier: BASIC | Application Surface: CANONICAL

Created At: 2026-09-09T19:29:42.370Z

VERDICT SUMMARY:

HIGH REGULATORY RISK (84/100)

Release Decision: BLOCKED | Verification Status: PARTIALLY_VERIFIED

STOP-SELL DIRECTIVE: CRITICAL_RISK_THRESHOLD: Score exceeds maximum institutional tolerance

Confidence Score: 99/100 Evidence Coverage: 98%

Merkle Root: sha256:62f8a8a24e70dd701bef355ab1060048c0c437d9d1d9ead54220f664e6e000e8

CRITICAL NOTICE: Contract is designated on international OFAC sanctions lists. Direct interaction triggers immediate automated compliance freezing across regulated institutions.

AUDIT OVERVIEW & CONTRACT CONTEXT

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Summary of execution scope and target objectives

Project / Contract: Tornado Cash Router
Contract Address: 0xd90e2f925da726b50c4ed8d0fb90ad053324f31b
Network / Chain ID: Ethereum Mainnet (ID: 1)
Token / Symbol: TORN-ROUTER
Proxy Pattern: Immutable (No Proxy)
Compiler Version: solc 0.7.6

SOURCE CODE VERIFICATION & BYTECODE DECOMPILEATION

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

State hash match against blockchain node and dispatcher analysis

EVM machine bytecode was disassembled into a control-flow graph (CFG). Standard ERC selectors and privileged opcodes were formally analyzed.

BASELINE SECURITY FINDINGS

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Identified vulnerabilities in baseline scan

Findings & Vulnerability Analysis:

CRITICAL VLM-TORN-01: OFAC Sanctions List Collision

Category: Regulatory Compliance | State: recommended

Recommendation: Institutional entities must block routing through this contract to maintain regulatory compliance.

PERMISSION & ROLE GOVERNANCE MAP (PRO) PRO

[LOCKED] [LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- Admin role: privilege and access control hierarchy in bytecode
- Verification of timelock delay parameters and governance controls
- Detection of privileged mint, pause, blacklist, and drainage capabilities
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

Protected evidence and findings withheld pending tier upgrade.

LIQUIDITY HOLDER CONCENTRATION & LOCK EVIDENCE (PRO) PRO

[LOCKED] [LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- Token distribution curve and top holder concentration analysis
- DEX liquidity pool pairing and lock verification across registered vaults
- Automated honeypot simulation, transfer fee checks, and sell restriction audit
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.

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ATTACK SURFACE & REENTRANCY FORMAL VECTORS (PRO)		PRO
[LOCKED]	[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]	

Summary of Analytical Depth in this section:

- Cross-function and cross-contract reentrancy guard modeling
- Price oracle staleness, manipulation resilience, and market feed verification
- Economic shock modeling and flash-loan manipulation resistance simulation
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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STORAGE LAYOUT COLLISION & MULTICOMPILER BYTECODE DIFF (ADVANCED)		ADVANCED
[LOCKED]	[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]	

Summary of Analytical Depth in this section:

- Storage slot allocation collision detection across implementation upgrades
- Upgrade proxy layout gap verification and state preservation checks
- Bytecode reproduction against official compiler metadata and build pipelines
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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FORMAL SMT INVARIANT PROOFS & REGRESSION ANALYSIS (ADVANCED)		ADVANCED
[LOCKED]	[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]	

Summary of Analytical Depth in this section:

- SMT proofs for solvency, conservation, and reentrancy impossibility invariants
- Bytecode regression analysis and logic change tracking
- Formal SMT Verification (Z3): PROPERTY P VERIFIED UNDER ASSUMPTIONS A FOR MODEL M
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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HUMAN ANALYST REVIEW & SIGNED VERIFICATION EVIDENCE (ADVANCED)		ADVANCED
[LOCKED]	[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]	

Summary of Analytical Depth in this section:

- Analyst sign-off requires verified reviewer evidence
- Cryptographically signed attestation digest
- Manual review boundary explicit: no certified safe claims
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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